
3AL Tutorial 4 2024

1. Let $s = \text{lcm}(m, n)$. Use the First Isomorphism Theorem to prove that $\mathbb{Z}_s$ is isomorphic to a subgroup of $\mathbb{Z}_m \times \mathbb{Z}_n$.

HINT: Recall that $\mathbb{Z}_n \cong \mathbb{Z}/n\mathbb{Z}$.

2. Let H and K be normal subgroups of a group G such that $G = HK$.

- (a) Prove that $G/(H \cap K) \cong (G/H) \times (G/K)$.

HINT: Use the First Isomorphism Theorem. To prove surjectivity, note that for any $g_1, g_2 \in G$, we have $g_1^{-1}g_2 \in G = HK$. So $g_1^{-1}g_2 = hk$ for some $h \in H$ and $k \in K$. Consider the element $x = g_2k^{-1} = g_1h \in G$.

- (b) Prove that if H and K are disjoint, then $G/K \cong H$ and $G/H \cong K$.

NOTE: Two subgroups are said to be disjoint if their intersection is the trivial subgroup $\{1\}$.

NOTE: We will use these results later in the course.

3. Let K be a normal subgroup of a finite group G such that K has prime index p in G . Prove that for all subgroups H of G either:

- (i) $H \subseteq K$, or,
(ii) $G = HK$ and $|H : H \cap K| = p$.

4. Let G be a group with normal subgroups H and K such that $H \subseteq K$. Prove that if G/H is cyclic, then so are G/K and K/H .

5. Let a and b be positive integers such that $a \mid b$. Determine the order of the factor group $a\mathbb{Z}/b\mathbb{Z}$.

Extra problems

Recall that for subgroup H and K of a group, we have the product HK . In an additive group, we typically write the product as $H + K = \{h + k : h \in H, k \in K\}$.

Consider the additive group $\mathbb{Z}$ and let $a, b \in \mathbb{Z}^+$.

(I) Prove that $a\mathbb{Z} + b\mathbb{Z} = \gcd(a, b)\mathbb{Z}$.

HINT: For all $a, b \in \mathbb{Z}^+$, there exist $n, m \in \mathbb{Z}$ such that $\gcd(a, b) = an + bm$.

(II) Prove that $a\mathbb{Z} \cap b\mathbb{Z} = \text{lcm}(a, b)\mathbb{Z}$.

(III) Use the Second Isomorphism Theorem to show that $ab = \gcd(a, b) \text{lcm}(a, b)$.